

Research on WSN coverage optimization based on Improved PSO Algorithm

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ABSTRACT

When PSO algorithm is applied to coverage optimization in wireless sensor networks, there will be some problems, such as slow convergence and premature phenomenon. To solve these problems, this paper proposes a wireless sensor network optimization algorithm based on standard PSO algorithm. The algorithm effectively improves the coverage rate of WSN network by improving the inertia weight and learning factor. Through simulation and analysis of the optimized coverage effect and convergence of coverage rate, the results show that the improved algorithm is better than the standard PSO algorithm and the improved algorithm in literature [4] in convergence speed and accuracy, and improves the problems of premature and slow convergence.

CCS CONCEPTS

• **Applied computing** → Computer forensics; Network forensics.

KEYWORDS

Wireless sensor network, Coverage optimization, PSO

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1 INTRODUCTION

With the development of computer technology, wireless sensor networks are more and more extensively used in military and civil applications. However, due to the limited energy of sensor nodes, the coverage and energy consumption of sensor networks have become the hot spots of wireless sensor research. The arrangement of sensing nodes generally adopts the method of random dripping from the air, which causes randomness in the deployment of nodes. Therefore, coverage becomes a major problem in the establishment of the whole network.

In recent years, many domestic and foreign scholars have done a lot of research and improvement on the coverage optimization

strategy of WSN networks. Ren et al. [1] discussed the coverage control algorithm and protocol in network deployment, and deeply analyzed the urgent problems of coverage control in wireless sensor networks. Guo et al. [2] summarized the research status of WSN deployment algorithm, and gave the application key points and future development trend of WSN deployment algorithm. To resolve the problem of low coverage caused by high aggregation of wireless sensor network nodes, an improved gray wolf optimization algorithm was proposed and applied to the optimal coverage of wireless sensor network nodes by Hu et al. [3]. To improve the coverage efficiency of wireless sensor network and the coverage rate of sensor nodes in the target area, Wang et al. [5] used the water wave optimization algorithm to optimize the deployment coordinates of sensor nodes.

To maximize the coverage of WSN, a PSO algorithm based on improved inertia weight and learning factor is invoked to optimize the coverage of WSN sensor nodes. The simulation experiment contrasts with the standard PSO algorithm to verify the effectiveness of the improved strategy, and then compared with the algorithm in literature 4 under the same conditions to verify the application value of the algorithm [6, 7].

2 WSN COVERAGE MODEL

It is assumed that the target surveying area is two-dimensional, the number of sensor nodes with the same structure placed in the area is N , the position coordinate value of each node is assumed to have been initialized, and the sensing radius of the node is r . The sensor node set is represented as $\text{Node}\{x_1, x_2, \dots, x_N\}$, $\text{node}_i = \{x_i, y_i, r\}$ refers to the circle, the center of which is node (x_i, y_i) and the monitoring radius of which is r . Suppose that the surveying area is divided into $l * k$ pixels, the coordinate of the pixel is (x, y) , and the distance between the pixel and sensor node is shown in formula 1.

$$d(\text{node}_i, p) = \sqrt{(x_i - x)^2 + (y_i - y)^2} \quad (1)$$

The events which the pixels covered with sensor nodes in target region are defined as c_i , the probability of occurrence of the event is p_{c_i} , that is the probability that the pixel covered with sensor node is shown in formula 2.

$$P_{\text{cov}}(x, y, \text{node}_i) = \begin{cases} 1, & \text{if } d(\text{node}_i, p) \leq r \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

The area coverage of sensor nodes in target surveying environment is defined as coverage ratio of sensor node set to the region of the surveying area, which is calculated by formula 3.

$$R_{\text{area}}(\text{node}) = \frac{\sum P_{\text{cov}}}{\text{Area}} \quad (3)$$

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Our ultimate goal is to find a group of nodes to maximize the coverage.

3 PSO ALGORITHM AND ITS IMPROVEMENT STRATEGY

3.1 Basic PSO

PSO is a group optimization algorithm inspired by the predatory behavior of birds. It focuses on the position and velocity attributes of particles. Each particle searches independently in the target space. They remember the optimal solution they found, and they also know the optimal solution currently found by the whole particle swarm. Where to go next depends on the current direction of the particles, the direction of the optimal solution they have found, and the direction of the current optimal solution of the whole particle swarm [8-10].

Suppose there are N particles $X = (X_1, X_2, \dots, X_n)$, which are optimized in the n -dimensional space. The position vector of the i th particle is $X_i = (X_{i1}, X_{i2}, \dots, X_{in})^T$ and the speed is $V_i = (V_{i1}, V_{i2}, \dots, V_{in})^T$. The corresponding fitness value is calculated according to the fitness function, and the global optimal solution $P_g = (P_{g1}, P_{g2}, \dots, P_{gd})^T$ is updated. The speed and position of the particle are updated using formulas 4 and 5.

$$V_{id}^{k+1} = \omega V_{id}^k + c_1 r_1 (P_{id}^k - X_{id}^k) + c_2 r_2 (P_{gd}^k - X_{id}^k) \quad (4)$$

$$X_{id}^{k+1} = X_{id}^k + V_{id}^{k+1} \quad (5)$$

In the formulas, ω is inertia weight, $d = 1, 2, \dots, n$, k is the current iteration number, V_{id} is the velocity of particles, c_1 and c_2 are learning factor, r_1 and r_2 are random number distributed in $[0,1]$ interval. The position is usually in section $[-X_{\max}, X_{\max}]$, and speed is in section $[-V_{\max}, V_{\max}]$.

3.2 Improved PSO

There are three important parameters in velocity iteration formula of PSO algorithm: inertia weight ω and learning factors c_1 and c_2 . ω is the inertia weight, which determines the influence of the particle's historical flight speed on the current flight speed. The selection of inertia weight is of great significance to the search ability of balance algorithm. In the iterative process, the global and local properties of the algorithm should be considered, and appropriate inertia weight should be adopted to search. In this paper, formula (6) is used to update the inertia weight.

$$\omega = \omega_{\max} + (\omega_{\max} - \omega_{\min}) * (t/t_{\max} - (t/t_{\max})^2) \quad (6)$$

In the formulas, t is current iterations, $t_{\max}$ is the maximum size of iterations, generally $\omega_{\max}=0.9$, $\omega_{\min}=0.4$.

c_1 is the embodiment of particles' self-learning and summarizing ability, and represents individual self cognition; Learning factor c_2 is the embodiment of the ability of particles to learn from better performing particles, and represents the social cognition of particles. appropriate c_1 and c_2 need to be set to facilitate particle search. At the initial stage, we should pay attention to the ability of individual self-awareness, and at the later stage, we should pay attention to the ability of individual to obtain social information. The learning factors are set using formulas 7 and 8.

$$c_1 = X + Z * \cos(\pi * \omega) \quad (7)$$

$$c_2 = Y - Z * \cos(\pi * \omega) \quad (8)$$

In the formulas, X , Y and Z is constant, by testing the typical test functions such as Griewank, sphere and Rosenbrock and using the existing learning factor adjustment strategies for reference, a group of coefficient combinations are obtained: $X=1.3$, $Y=2.0$, $Z=1.2$.

4 WSN COVERAGE OPTIMIZATION BASED ON IMPROVED PSO ALGORITHM

The optimization objective of this improved algorithm is to maximize the WSN network coverage rate (formula 3), and output the optimized network coverage rate and the coordinates of all sensor nodes.

4.1 Algorithm steps

The steps of algorithm proposed in this paper are as follows:

Step 1, Set the area to be covered and the number of target nodes, generate the sensor node set according to the number of target nodes, and use the area coverage as the fitness function.

Step 2, Initialize the position, velocity, maximum iteration times $t_{\max}$ and other parameter conditions of all particle, and calculate local optimal value and global optimal value of all particle according to formula 1-3.

Step 3, Update inertia weight ω and two learning factors according to formula 6-8.

Step 4, Update the position and speed of the particles, and calculate fitness value of every particle separately again. Compare the fitness value with local optimal value, and judge whether to modify local optimal value on the basis of comparison results. When a new local optimal value is found, compare it with the previous global optimal value, and judge whether to modify the global optimal value judging by the comparison results.

Step 5, Judge whether t is equal to the maximum number of iterations $t_{\max}$. if not, $t = t + 1$, returns to step 2. If it is equal to $t_{\max}$, the global optimal function value (maximum coverage rate) and its vector (coordinates of all sensor nodes) are output, and the algorithm ends.

5 SIMULATION AND ANALYSIS

5.1 Simulation platform and parameter setting

The running environment of the experiment is win7, i5 3.20Ghz processor, 8g memory, and all algorithm codes are programmed with matlab 2017b.

Suppose 35 wireless nodes are arranged in a region of 50m×50m, the sensing semidiameter of all sensor nodes is 3m, the communication semidiameter is 6m, the population size is 20, and maximum iterations is 500, $\omega_{\max}=0.9$, $\omega_{\min}=0.4$.

In order to compare the performance of WSN coverage optimized by the algorithm in this paper, we choose to compare and simulate with the basic PSO and the value taking strategy II of the improved PSO algorithm proposed in literature [4]. Each algorithm runs 50 times separately, and the experimental results of each time are recorded. The values of ω , c_1 and c_2 in the three algorithms are shown in Table 1.

Table 1: Parameters setting of PSO and its variants

algorithm	inertia weight factor ω	learning factor c_1	learning factor c_2
basic PSO algorithm	$\omega = \omega_{\max} - ((\omega_{\max} - \omega_{\min})/t_{\max}) * t$	2	2
algorithm of literature [4]	$\omega = \omega_{\min} + (\omega_{\max} - \omega_{\min}) * \exp(-2 * (t/t_{\max})^6)$	$c_1 = 0.5 * \omega + 1.5$	$c_2 = 2.5 - c_1$
algorithm of this paper	$\omega = \omega_{\max} + (\omega_{\max} - \omega_{\min}) * (t/t_{\max} - (t/t_{\max})^2)$	$c_1 = X + Z * \cos(\pi * \omega)$	$c_2 = Y - Z * \cos(\pi * \omega)$

Table 2: Experimental results of WSN coverage of the three algorithms

indicators	basic PSO algorithm	algorithm of literature [4]	algorithm of this paper
Ave	0.917454825	0.871318724	0.969300269
Std	0.030529952	0.015452433	0.007247976
Med	0.921184	0.87005	0.970973
Best	0.973472	0.88466	0.990388
Worst	0.815071	0.856209	0.951942

5.2 Performance evaluation index of the algorithm

The optimization performance of the three algorithms was evaluated by average (Ave), standard deviation (Std), median (Med), Best (Best), Worst (Worst) and box graph.

(1) Ave reflects the average performance of the algorithm, which uses formula 9 to calculate.

$$\text{Ave} = \frac{1}{m} \sum_{i=1}^m \text{value}_i \quad (9)$$

In the formula, m is the number of times the algorithm is run, and value_i is the running result of the algorithm when it runs for the i th time.

(2) Std reflects the stability of the algorithm, which uses formula 10 to calculate.

$$\text{Std} = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (\text{value}_i - \text{Ave})^2} \quad (10)$$

In the formula, value_i is the running result of the algorithm when it runs for the i th time, Ave is the average value of the algorithm, and m is the number of algorithm runs.

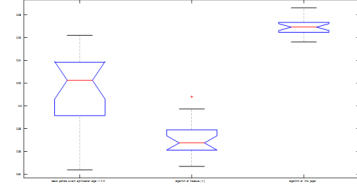
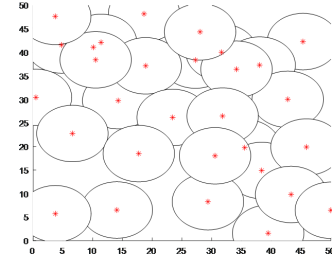
(3) Med reflects the distribution of fitness values of the algorithm.

(4) Best reflects the fitness value of the algorithm obtained by m times of running.

(5) Worst reflects the Worst value obtained by m times of running of the algorithm's fitness value.

5.3 Effect of sensor node coverage

To improve the reliability of simulation results, 50 simulations were conducted under the same environment. The Ave, Std, Med, Best and Worst values of the three algorithms are shown in Table 2. The box diagram of WSN coverage fitness function optimized by the three algorithms is shown in Figure 1. The coverage effect optimized by the basic PSO is shown in Figure 2, the coverage effect optimized by the improved PSO algorithm proposed in literature [4] is shown in Figure 3, and the coverage effect optimized by the algorithm in

**Figure 1: Box diagram of WSN coverage by three algorithms****Figure 2: coverage effect of basic PSO algorithm**

this paper is shown in Figure 4. In the figure, the center of the circle is the deployment coordinate of the node, and the arc is sensing range of the sensor.

5.4 Result Analysis

After 50 simulations, Simulation results is shown in Figure 5. which shows that the coverage of the basic PSO algorithm converges to 91.7%, the improved PSO algorithm proposed in reference [4] converges to 86.91%, and the coverage of the algorithm in this paper converges to 97.04%. The number of iterations of convergence is about 250.

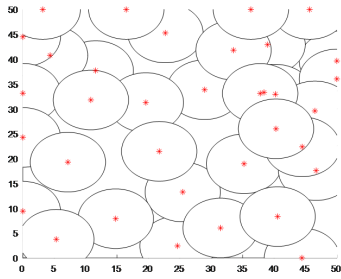


Figure 3: coverage effect of literature [4] algorithm

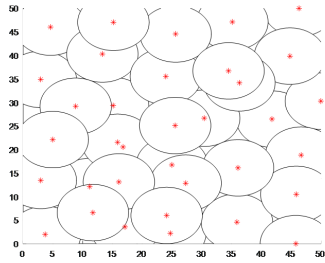


Figure 4: coverage effect of this paper algorithm

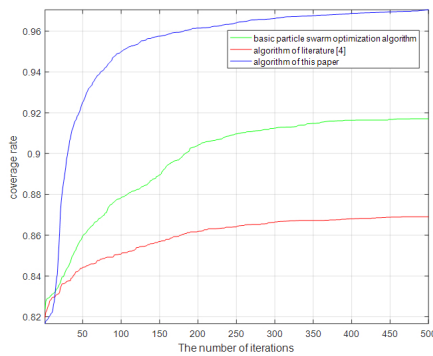


Figure 5: Fitness curves of the three algorithms

6 CONCLUSION

Aiming at the shortcomings of uneven node distribution and low coverage in random deployment mode of sensor network, an improved PSO method was mooted for WSN node optimization. The convergence speed of PSO can be improved and the convergence precision can be further increased by using the adaptive weight. The algorithm in this paper is compared with the basic PSO algorithm and the improved PSO algorithm in literature 4. The three algorithms were run 50 times in the same environment, and the robustness of the algorithm was evaluated by six evaluation indexes. The results reflect that the improved PSO algorithm has better searching ability and convergence accuracy, and has stronger stability. Compared with the other two algorithms, the improved PSO

algorithm in this paper effectively improves the coverage of WSN nodes, thus reducing the configuration cost of the network. In the future, the convergence of the improved PSO algorithm will be further proved and studied, and it will be applied to the heterogeneous node deployment and Three-dimensional deployment problems of WSN.

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